

Asia Quant Academy

Lecture 7 - Foundation Finance I

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Introduction to Financial Market

Brief History of Financial Markets

Financial Market

Facilitate the trade of goods, services, and financial instruments.

Money

A legal tender for the benchmarking of the value of goods and services.

Market

A centralized places of trade, where supply and demand determine prices.

Modern-day Financial Market

Globalized, regulated, and digitized, with various functions and players operating within the system.

Functions, Role Players and Types

Functions	Role Players	Market types
• To raise capital • Profit incentive • Management of risk	• Government • Central banks • Investment banks • Insurance & re-insurance companies • Clearing houses • Dealers, intermediaries, brokers	• Money markets • Capital markets • Currency markets • Commodities markets • Other markets

Element of Financial Market - Part I

Financial markets constitutes different markets to trade different types of products. Below are some of the most common markets that one would look at:

- Debt market versus equity
- Primary markets versus secondary markets
- Exchange markets versus Over-the-counter (OTC)
- Spot/Cash and future markets

Element of Financial Market - Part II

For any of the market, there are some fundamental properties and mechanisms that one would be interested in, which are also important for quants

- Price determination and discovery
- Information aggregation and coordination
- Risk Sharing
- Liquidity
- Efficiency

Risk Management in Financial Market - Part I

- **Avoidance** - Avoidance, as a risk management technique, involves seeking risk-free investments like US Treasury Bills to mitigate default risk but does not protect against inflation risk.
- **Loss prevention or reduction** - Loss prevention and reduction, a risk mitigation technique, involves accepting risk while taking steps to minimize potential losses. Hedging, such as using futures or forward contracts, is an example of this technique.
- **Diversification** - Diversification, a risk management technique, involves pooling assets with different risk weightings to minimize potential losses and align with investors' risk appetite and return goals.
- **Transferring risk** - Risk mitigation techniques involve transferring risk from one party to another through premiums or other means, but each approach has its own inherent cost.

Risk Management in Financial Market - Part II

- **Money markets**

Short-term debt, highly liquid, associated with short-term interest.

Risk: Default risk

- **Capital markets**

Debt (bonds, debentures) and equity (shares), company performance affects share value.

Risk: prolonged exposure to potentially adverse price or payment risk

- **Currency markets**

Trading of international currencies, highly liquid, largest market in terms of daily trading volume.

Risk: political, economic and social risk

- **Commodity markets**

Trading of soft (coffee, wheat) and hard goods (iron ore, gold), characterized by derivative and futures trading.

Risk: prices move in an adverse direction to the futures trade

Time Value of Money

What is an asset?

- Business entity
- Property, plant, and equipment,
- Patents, R & D,
- Stocks, bonds, options, ...
- Knowledge, reputation, opportunities, etc.

From a business perspective, an asset is a sequence of cashflows.

Definition

$$\text{Asset}_t \equiv \{CF_t, CF_{t+1}, CF_{t+2}, \dots\}$$

Cashflows and Assets

Valuing a sequence of cashflows

Sequences of cashflows are the "basic building blocks of finance"

Definition

Value of $Asset_t \equiv V_t(CF_t, CF_{t+1}, CF_{t+2}, \dots)$

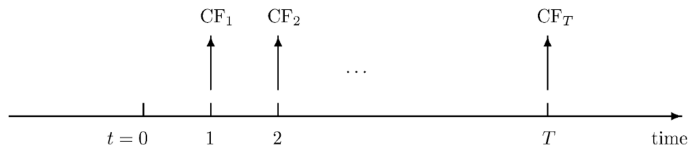


Figure: A series of cashflow at different time.

Present Value Operator

Consider manipulating foreign currencies:

$$¥150 + £300 \stackrel{?}{=} ??450$$

They cannot be added together without first converting into a common currency.

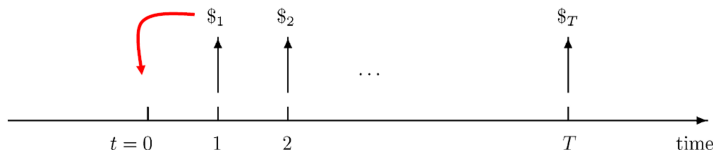
$$\begin{aligned} ¥150 + (£300) \times (153 ¥ / £) &= ¥46,050.00 \\ (¥150) \times (0.0065 £ / ¥) + £300 &= £ 300.98 \end{aligned}$$

What about cashflows of different dates?

Present Value Operator

Cashflows at different dates are different "Currencies".

- Past and future cannot be combined without first converting them.
- Once "exchange rates" are given, combining cashflows is trivial.



Definition

Present Value:
$$V_0(CF_1, CF_2, CF_3, \dots) = \frac{\$1}{\$0} \times CF_1 + \frac{\$2}{\$0} \times CF_2 + \dots$$

Present Value Operator

Initial cost of investment can be captured by the cashflow CF_0

Definition

$$\text{PV: } V_0(CF_0, CF_1, CF_2, CF_3, \dots) = CF_0 + \frac{\$1}{\$0} \times CF_1 + \frac{\$2}{\$0} \times CF_2 + \dots$$

Beware that any of the CF_i can be negative (i.e. future costs.)

Present Value Operator

Suppose we have the following "exchange rates"

$$\left(\frac{\$1}{\$0}\right) = 0.90, \quad \left(\frac{\$2}{\$0}\right) = 0.80$$

The net present value of a project requiring a current investment of \$10MM with cashflows of \$5MM in Year 1 and \$7MM in year 2:

$$NPV_0 = -\$10 + \$5 \times 0.90 + \$6 \times 0.80 = \$0.10$$

Time Value of Money

What determines the growth of \$1 over T years?

\$1 today should be worth more than \$1 in the future. Consider the opportunity cost of capital be r ,

$$\text{\$1 in Year 0} = \text{\$1} \times (1 + r) \text{ in Year 1}$$

$$\text{\$1 in Year 0} = \text{\$1} \times (1 + r)^2 \text{ in Year 2}$$

...

$$\text{\$1 in Year 0} = \text{\$1} \times (1 + r)^T \text{ in Year } T$$

Therefore,

$$V_0 = CF_0 + \sum_i \frac{CF_i}{(1 + r)^i}$$

Time Value of Money - Example

Your firm spends \$800,000 annually for electricity at its Boston headquarters. Johnson Controls offers to install a new computer-controlled lighting system that will reduce electric bills by \$90,000 in each of the next three years.

If the system costs \$230,000 fully installed, is this a good investment?

Lighting System*

Year	0	1	2	3
Cashflow	-230,000	90,000	90,000	90,000

Assume the cost savings are known with certainty and the interest rate is 4%.

Time Value of Money - Example

Lighting System

Year	0	1	2	3
Cashflow	-230,000	90,000	90,000	90,000
÷		1.04	$(1.04)^2$	$(1.04)^3$
PV	-230,000	86,538	83,210	80,010

$$\text{NPV} = -230,000 + 86,538 + 83,210 + 80,010 = \$19,758$$

Compounding

Interest may be credited/charged more often than annually:

- Bank accounts: daily
- Mortgages and leases: monthly
- Bonds: Semiannually

Effective annual rate may differ from annual percentage rate. Why?

Typical compounding conventions:

Let r denote *APR*, n periods of compounding.

$$r_{EAR} \equiv (1 + r/n)^n - 1$$

10% Compounded Annually, Semi-Annually, Quarterly, and Monthly

Month	\$1,000	\$1,000	\$1,000	\$1,000
1				\$1,008
2				\$1,017
3			\$1,025	\$1,025
4				\$1,034
5				\$1,042
6		\$1,050	\$1,051	\$1,051
7				\$1,060
8				\$1,069
9			\$1,077	\$1,078
10				\$1,087
11				\$1,096
12	\$1,100	\$1,103	\$1,104	\$1,105

Figure: Same APR, different EAR

Inflation

What is inflation?

- Change in real purchasing power of \$1 overtime
- Different from time-value of money
- For some countries, inflation is extremely problematic.

How to quantify its effects?

Wealth $W_t \Leftrightarrow$ Price Index I_t

Wealth $W_{t+k} \Leftrightarrow$ Price Index I_{t+k}

Increase in Cost of Living $\equiv I_{t+k}/I_t = (1 + \pi)^k$

"Real Wealth" $\widetilde{W}_{t+k} \equiv W_{t+k}/(1 + \pi)^k$

Definition

(Real Inflation) $r_{real} \approx r_{nominal} - \pi$

Proof:

"Real Wealth" $\widetilde{W}_{t+k} \equiv W_{t+k}/(1 + \pi)^k$

"Real Return" $(1 + r_{real})^k \equiv \frac{\widetilde{W}_{t+k}}{W_t}$

$$= \frac{W_{t+k}}{W_t} \frac{1}{(1 + \pi)^k} = \frac{(1 + r_{nominal})^k}{(1 + \pi)^k}$$
$$r_{real} = \frac{1 + r_{nominal}}{1 + \pi} - 1$$
$$\approx r_{nominal} - \pi$$

Bond

Coupon Bond

Consider the following bond: In May 2010 the U.S. Treasury sold a bond with,

- A coupon rate of $2\frac{1}{8}\%$ and,
- a maturity date of May 31, 2015,
- a payment frequency of two a year, six months apart

This bond is called *"2 1/8s of May, 2015"*.

Cash Flow of the Bond

The unit for bond purchasing is \$1,000. Suppose that an investor purchases \$m face value of the bond, i.e., 1,000 units. The coupon payment is calculated according to

$$\frac{1}{2} \times 2\frac{1}{8}\% \times \$1,000,000 = \$10,625$$

Coupon Bond

Cash flows of the U.S. 2 1/8 from books of May 31, 2015.

	Coupon Payment	Principal Payment
Date		
"11/30/2010"	\$10, 625	
"5/31/2011"	\$10, 625	
"11/30/2011"	\$10, 625	
"5/31/2012"	\$10, 625	
"11/30/2012"	\$10, 625	
"5/31/2013"	\$10, 625	
"11/30/2013"	\$10, 625	
"5/31/2014"	\$10, 625	
"11/30/2014"	\$10, 625	
"5/31/2015"	\$10, 625	\$1, 000, 000

Coupon Bond

The price of coupon bonds are quoted as a percentage of the principal value, using a price tick of one-32nd of a percentage point:

$$101 - 12_{3/4} = 101 + \frac{12.75}{32} = 101.3984$$

More on quote prices:

- "+" means 0.5
- The actual tick size can be $\frac{1}{32} \times \frac{1}{8} = \frac{1}{256}$

So, we have:

$$108 - 31+ = 108 + \frac{31.5}{32} = 108.9843$$

$$102 - 18_{1/8} = 103 + \frac{18.125}{32} = 102.5664$$

TABLE 1.1 Selected Treasury Bond Prices for Settlement on February 15, 2001

Coupon	Maturity	Price
7.875%	8/15/01	101-12 $\frac{3}{4}$
14.250%	2/15/02	108-31+
6.375%	8/15/02	102-5
6.250%	2/15/03	102-18 $\frac{1}{8}$
5.250%	8/15/03	100-27

Accrued Interest

Denote, flat price by p , accrued interest by AI , present value for the cashflows by PV , and full price by P .

Definition

(Invoice price) $P = p + AI = PV$

Consider a bond with semi-annual coupon 3.625. For \$100 notional, the coupon value is $\frac{3.625}{2} = 1.8125$.

The accrued value from 15/2 to 6/1 is $\frac{106}{181} \times 1.8125 = 1.0615$.

Therefore, the transaction price of the bond per 100 face amount is $102.8125 + 1.0615 = 103.8740$.

For \$10,000 face amount, the invoice price is $100 \times \$103.8740 = \$10,387.40$.

Quote using Flat Price

If the bond is quoted using dirty price (i.e. invoice price), there will be a jump of price across a coupon date.

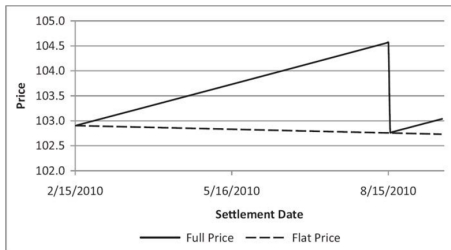


Figure: The full price jump across a coupon date

Market Jargons

Flat Price: Clean price, quote price

Full Price: Invoice price, transaction price, dirty price

Day Count Conventions

There are several conventions for day count:

- Actual/Actual: Government bonds
- Actual/360: The actual number of days between two dates divided by 360 (money markets, floating-leg swaps)
- 30/360: Assume 30 days in any month (corporate bonds and fixed-leg of swaps)

Yield-to-maturity (YTM)

Consider a semi-annual bond,

Definition

$$P = \frac{c}{2} \sum_{t=1}^{2T} \frac{1}{(1 + \frac{y}{2})^t} + \frac{1}{(1 + \frac{y}{2})^{2T}}$$

After doing some manipulation,

$$P = \frac{c}{y} \left(1 - \frac{1}{(1 + \frac{y}{2})^{2T}} \right) + \frac{1}{(1 + \frac{y}{2})^{2T}}$$

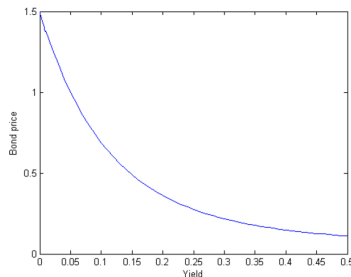


Figure: Price of a 10y 5s bond vs. yield

Yield-to-maturity (YTM)

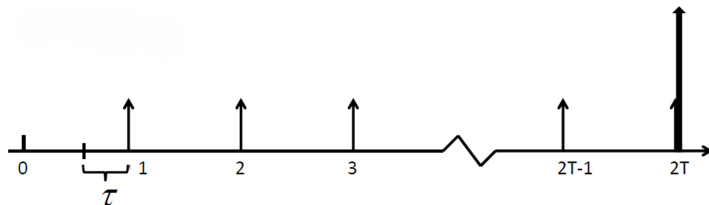
For example, the price of the 4 1/2s of November 30, 2011 was 105.856. The yield-to-maturity, y , of this bond is therefore defined such that:

$$105.856 \equiv \frac{2.25}{1 + \frac{y}{2}} + \frac{2.25}{(1 + \frac{y}{2})^2} + \frac{102.25}{(1 + \frac{y}{2})^3}$$

By using numeric solvers, one may show that the yield of the bond is about 0.574%.

General Bond Price Formula

When the first coupon is due in τ fraction of a half year,



The bond price formula becomes,

$$P = \left(1 + \frac{y}{2}\right)^{1-\tau} \frac{C}{y} \left(1 - \frac{1}{\left(1 + \frac{y}{2}\right)^{2T}}\right) + \frac{1}{\left(1 + \frac{y}{2}\right)^{2T}}$$

Discount Factors

Denote the discount factor for t years by $d(t)$.

Let $d(.5) = 0.99925$, then the PV of \$1 to be received in six months is 99.925 cents.

The present value of the cash flow \$1,050,000 in six month is $0.99925 \times \$1,050,000 = 1,049,213$.

Question: Can we bootstrap a discount factor curve from a set of coupon bonds?

Bootstrapping the Discount Factors

Lets imply the discount factor for each time period one by one.

$$100.55 = (100 + \frac{1\frac{1}{4}}{2})d(.5)$$

$$\rightarrow d(.5) = .99925$$

$$104.513 = \frac{4\frac{7}{8}}{2} \times d(.5) + (100 + \frac{4\frac{7}{8}}{2})d(1)$$

$$\rightarrow d(1) = .99648$$

$$105.856 = \dots + (100 + \frac{4\frac{1}{2}}{2})d(1.5)$$

$$\rightarrow d(1.5) = .99135$$

Coupon	Maturity	Price
1.250%	"11/30/2010"	100.55
4.8750%	"5/31/2011"	104.513
4.500%	"11/30/2011"	105.856
4.75%	"5/31/2012"	107.966
3.3750%	"11/30/2012"	105.869
3.500%	"5/31/2013"	106.76
2%	"11/30/2013"	101.552
2.25%	"5/31/2014"	101.936
2.1250%	"11/30/2014"	100.834

Figure: List of bonds for bootstrapping

Spot Rates

A spot rate is the rate on a spot loan, an agreement between a lender and a borrower at the time of the agreement. Denote the semiannually compounded t-year spot rate by $\hat{r}(t)$.

Definition

Discount Factor & Spot Rates:
$$d(t) = \frac{1}{\left(1 + \frac{\hat{r}(t)}{2}\right)^{2t}}$$

Macaulay Duration

Bond subject to interest-rate risk:

- As interest rates change, bond prices also change.
- Sensitivity of price to changes in yield measures risk.

Definition

(Macaulay Duration): $D_m = \sum_{k=1}^T k \cdot w_k$

where $w_k = \frac{C_k/(1+y)^k}{P} = \frac{PV(C_k)}{P}$

Sensitivity of bond prices to yield changes:

$$P = \sum_{k=1}^T \frac{C_k}{(1+y)^k}$$
$$\frac{\partial P}{\partial y} = \frac{-1}{1+y} \sum_{k=1}^T k \cdot \frac{C_k}{(1+y)^k}$$
$$\frac{1}{P} \frac{\partial P}{\partial y} = -\frac{D_m}{1+y} = -D_m^*$$

Macaulay Duration - Example

Consider a 4-year T-note with face value \$100 and 7% coupon, selling at \$103.5, yielding 6%. For T-notes, coupons are paid semi-annually. Using 6-month intervals, the coupon rate is 3.5% and the yield is 3%.

- Duration(in 1/2 year units) is $D = (738.28)/103.5 = 7.13$.
- Modified duration is $D^* = D/(1 + y) = 7.13/1.03 = 6.92$.
- Price risk at $y = 0.03$ is $\delta = D^* \times P = (6.92)(103.5) = 716$.

t	CF_t	$PV(CF_t)$	$t \cdot PV(CF_t)$
1	3.5	3.40	3.40
2	3.5	3.30	6.60
3	3.5	3.20	9.60
4	3.5	3.11	12.44
5	3.5	3.02	15.10
6	3.5	2.93	17.59
7	3.5	2.85	19.92
8	103.5	81.70	653.63
		103.50	738.28

Figure: Cash flow of the bond

Convexity

Sensitivity of duration to yield changes:

$$\frac{\partial^2 P}{\partial y^2} = \frac{1}{(1+y)^2} \sum_{k=1}^T k \cdot (1+k) \cdot \frac{C_k}{(1+y)^k}$$
$$\frac{1}{P} \frac{\partial^2 P}{\partial y^2} = V_m$$

Relation between duration and convexity:

$$\begin{aligned} P(y') &\approx P(y) + \frac{\partial P}{\partial y}(y) \cdot (y' - y) + \frac{\partial^2 P}{\partial y^2}(y) \cdot \frac{(y' - y)^2}{2} \\ &= P(y) \cdot [1 - D_m^*(y' - y) + \frac{1}{2} V_m (y' - y)^2] \end{aligned}$$

Convexity - Example

Breakdown by cash flows

$$D_m^* \equiv -\frac{1}{P} \frac{\partial P}{\partial y} = \sum_j \frac{P_j}{P} D_{m,j}^*$$

$$V_m^* \equiv -\frac{1}{P} \frac{\partial^2 P}{\partial y^2} = \sum_j \frac{P_j}{P} V_{m,j}^*$$

Example:

$$D_m^* = \frac{1}{1 + \frac{0.06}{2}} \sum_{k=1}^8 \frac{kC_k}{2P(1 + \frac{0.06}{2})^2} = 3.51$$

$$V_m^* = \frac{1}{(1 + \frac{0.06}{2})^2} \sum_{k=1}^8 \frac{k(k+1)C_k}{4P(1 + \frac{0.06}{2})^k} = 14.81$$

Numerical Example For Duration and Convexity
6% 4-Year Bond, Yield-to-Maturity = 6%, $P=100$

k	C_k	$\frac{C_k}{(1 + \frac{0.06}{2})^k}$	$\frac{kC_k}{2P(1 + \frac{0.06}{2})^k}$	$\frac{k(k+1)C_k}{4P(1 + \frac{0.06}{2})^k}$
1	3	2.912621	0.014563	0.014563
2	3	2.827787	0.028277	0.042416
3	3	2.745424	0.041181	0.082362
4	3	2.665461	0.053309	0.133273
5	3	2.587826	0.064695	0.194086
6	3	2.512452	0.075373	0.263807
7	3	2.439274	0.085374	0.341498
8	103	81.309151	3.252366	14.635647

Figure: Numerical example for duration and convexity